State-versus-County Aggregation and Identification in U.S. Monetary Policy Transmission to Employment, 2002–2019

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Abstract

A shift-share (Bartik) difference-in-differences design applied to U.S. county employment returns a *positive* response to federal-funds-rate tightening—the opposite of the contractionary effect predicted by theory and recovered by textbook vector autoregressions. This paper asks whether that reversal reflects spatial aggregation or *identification*. Using a county employment panel (2002–2019), Bauer–Swanson high-frequency monetary policy surprises, and FRED macro series, I estimate four time-series models: a reduced-form VAR, a recursively identified structural VAR, Jordà local projections, and a random forest. The reduced-form and structural VARs both deliver the conventional negative employment response on a pre-ZLB sample (trough $\approx -0.38\%$), while the same VAR estimated over 2002–2019 inverts because the zero lower bound renders the funds rate an uninformative policy indicator. Replacing the endogenous rate change with the identified high-frequency surprise in the shift-share design *flips the exposure interaction from a significant positive (*$+0.20$ *to* $+1.25$*) to negative (*-0.39 *to* -2.42*) across horizons*— evidence pointing to identification, rather than aggregation, as the driver, though the surprise-based estimates are imprecise. The random forest lowers RMSE relative to naïve benchmarks but has zero correlation with realized employment growth, indicating variance shrinkage rather than genuine predictability. The central methodological lesson: endogeneity of the policy instrument can manufacture a sign reversal in an otherwise standard regional design.

1 Introduction

A long tradition in regional macroeconomics holds that contractionary monetary policy reduces employment, and that interest-sensitive local economies—those concentrated in construction and durable manufacturing—contract *more* (Carlino and DeFina, 1998). A shift-share difference-in-differences (DiD) design I estimated in prior work reverses this prediction: interacting the quarterly change in the federal funds rate with predetermined 2002 construction-plus-manufacturing exposure across ∼2,900 U.S. counties yields a *positive, statistically significant* coefficient (+0.0202, s.e. 0.0045; cumulative four-quarter effect +0.042). Read literally, high-exposure counties *gain* employment when the Fed tightens—a result at odds with both theory and the conventional vector-autoregression (VAR) evidence.

This paper asks a precise question: **is the positive sign an artifact of spatial aggregation, or of identification?** The two explanations have very different implications. If aggregation drives it, the county design is measuring something real that disappears at coarser geographies. If identification drives it, the problem is that the realized funds-rate change is endogenous—it rises precisely when the economy (and interest-sensitive sectors) is booming—so the "effect" is reverse causation, and replacing the endogenous rate with an exogenous monetary *shock* should restore the contractionary sign.

The gap I address is that regional transmission studies rarely confront the two explanations head-on within a single, internally consistent research design. I do so by holding the shift-share structure fixed and varying only the source of policy variation, while benchmarking against national VAR/SVAR systems and a machine-learning forecaster. Concretely, I estimate four of the methods on the ECO 508 list: a reduced-form **VAR**, a recursively identified **SVAR**, Jordà **local projections** (LP) using Bauer–Swanson high-frequency surprises as the identified shock, and a **random forest** (RF) for employment-growth forecasting. The full-panel DiD is retained only as motivating evidence, not re-estimated as a headline result.

Three findings emerge. First, the reduced-form and structural VARs deliver the textbook negative employment response on a pre-ZLB sample (cumulative trough $\approx -0.38\%$), but the *same* specification estimated over 2002–2019 inverts to a positive response—because the zero-lower-bound (ZLB) episode of 2008–2015 pins the funds rate near zero and destroys its content as a policy measure. Second, and centrally, when I replace the endogenous rate change with the identified surprise inside the shift-share design, the exposure interaction *flips sign* from significantly positive to negative at every horizon, though the surprise-based estimates are not individually significant given limited high-frequency variation. Third, the random forest attains lower out-of-sample RMSE than naïve benchmarks but has essentially zero correlation with realized employment growth, a cautionary illustration of variance shrinkage masquerading as skill. The remainder of the paper reviews the relevant literature (§2), describes the data (§3) and methodology (§4), presents results (§5), and concludes (§6).

2 Literature Review with Paper Summaries

The empirical benchmark for this paper is Carlino and DeFina (1998), who ask whether U.S. regions respond differently to a common monetary policy shock and whether structural features explain the heterogeneity. They estimate a structural VAR in output, prices, and the funds rate across eight BEA regions over 1958–1992, tracing regional personal-income responses to a funds-rate innovation. A core of regions tracks the national average, but the durable-manufacturing-heavy Great Lakes contracts more sharply while the Rocky Mountains and Southwest contract less, with the gaps lining up with industry mix and small-firm shares. Theirs is the canonical statement of the prediction my difference-in-differences design violates: greater interest-sensitivity should mean a *larger* contraction, not an employment gain, so a positive county response signals that identification, rather than a genuine regional regularity, is at work.

Coibion (2012) asks why the estimated size of monetary effects is so method-sensitive, reconciling the small effects implied by recursive VARs with the large effects implied by the Romer–Romer narrative approach. Working with postwar U.S. aggregates, he shows that once the contractionary impetus, the reserves-targeting period, and lag length are accounted for, the real effects are consistent across schemes and "medium"-sized. The relevance for my project is twofold: it legitimizes placing VAR and local-projection responses on a common scale, and it warns that sample period and lag choice can swing conclusions—a warning that proves central to my zero-lower-bound result.

On identification, Christiano et al. (1999) supply the template for my structural VAR. They ask what the dynamic effects of a monetary shock are and how sensitive those effects are to identifying assumptions, using recursive (Cholesky) identification on U.S. aggregates with the funds rate ordered after slow-moving output and prices. A contractionary shock produces a hump-shaped decline in output and employment that peaks after one to two years, alongside the familiar "price puzzle." That timing restriction and the expected hump-shaped contraction are exactly what my benchmark SVAR reproduces.

My primary estimation vehicle is the local projection of Jordà (2005), motivated by the goal of estimating impulse responses without committing to a full dynamic system. The method regresses the outcome at each horizon directly on the shock—one regression per horizon—with heteroskedasticity- and autocorrelation-consistent standard errors. Jordà shows these responses are robust to dynamic misspecification and nest VAR responses as a special case, at some cost in efficiency. This is what lets me trace the county employment response to an external monetary shock horizon by horizon rather than imposing a panel VAR.

Plagborg-Møller and Wolf (2021) then ask whether that choice of estimator matters, proving that local projections and VARs estimate the same population impulse responses given comparable lag structures. The result is theoretical, with finite-sample simulations showing the two differ only through a bias–variance trade-off. For my paper this means any divergence between the LP and the VAR/SVAR reflects finite-sample and shock-measurement differences, not the estimator—so the LP-versus-VAR comparison can be read as a consistency check rather than a contest.

The shock that feeds those projections comes from the high-frequency identification literature. Gürkaynak et al. (2005) ask whether asset-price responses to FOMC announcements can isolate unanticipated policy, using thirty-minute changes in money-market futures around announcements. They find that two factors are needed—a current-rate surprise and a path surprise—establishing that narrow-window futures changes capture the component of policy markets did not expect. This is the methodological foundation for treating a high-frequency surprise as an exogenous shock.

Nakamura and Steinsson (2018) use such surprises to measure monetary non-neutrality and to test for a "Fed information effect," studying real-rate and expectations responses in thirty-minute windows around announcements. They find sizable real effects but also that surprises move expected output growth in a direction suggesting announcements reveal the central bank's private read on the economy. Their information effect is the leading alternative explanation for perverse-signed responses, which I carry explicitly into my limitations.

Bauer and Swanson (2023), whose orthogonalized monthly surprise series I adopt, ask whether high-frequency surprises are truly exogenous given evidence they are predictable. Using an expanded announcement sample, they show raw surprises are partly forecastable from pre-announcement macro-financial conditions and recommend orthogonalizing them. I use their cleaned series, and I document that it is essentially uncorrelated with the *realized* quarterly funds-rate change—the reason an instrumental-variables projection is infeasible here and the surprise must enter directly as the shock.

Gertler and Karadi (2015) formalize how such surprises identify structural responses as external instruments, asking what monetary policy does to activity and credit costs in a proxy-SVAR instrumented by funds-futures surprises. They find that modest movements in short rates generate large movements in credit spreads alongside a textbook output contraction, while Stock and Watson (2018) give the general conditions under which this external-instrument approach (equivalently, LP–IV) is valid. This is the design I first attempt, abandoning the instrumental-variables version only once its quarterly first stage proves degenerate.

Jarociński and Karadi (2020) push the information-effect concern furthest, asking whether surprises can be decomposed into pure monetary and central-bank information shocks. Within a structural VAR they use the sign of the co-movement between interest rates and stock prices around announcements to separate the two: pure tightenings produce textbook contractions, information shocks produce opposite-signed responses, and conventional surprises blend them. This is the sharpest alternative reading of my puzzle—an exposure design built on contaminated surprises could itself generate a perverse positive response—so I flag it as the most important unresolved threat.

Because my design is itself a shift-share (Bartik) object, its identification deserves equal scrutiny. Goldsmith-Pinkham et al. (2020) ask precisely what a shift-share instrument identifies, showing analytically that it is numerically equivalent to using the predetermined exposure shares as instruments. Identification therefore rests on the exogeneity of the shares and is driven largely by the cross-section. This disciplines how I read the exposure interaction and points to the shares—and the endogeneity of the shock interacted with them—as the threat to identification.

Borusyak et al. (2022) develop the complementary view, asking when identification can instead rest on the *shocks*. They show shift-share IV is valid under many quasi-randomly assigned shocks, provided no small set of shocks dominates. Both warnings bite in my setting: with few and endogenous monetary "shifts" and exposure shares plausibly correlated with county characteristics, a shift-share design is exactly the configuration in which an endogenous policy rate can flip a coefficient's sign.

The machine-learning component is motivated by Medeiros et al. (2021), who ask whether machine-learning methods improve macroeconomic forecasting in a data-rich environment. Using a large U.S. macro dataset and rolling out-of-sample evaluation, they find random forests outperform factor and autoregressive benchmarks for inflation, owing to data-driven variable selection and nonlinearity. I adopt their lagged-feature, rolling-window protocol; my own finding—that the forest lowers RMSE for employment growth yet carries no directional skill—serves as a counterpoint about when such methods genuinely help.

Taken together, the applied regional work fixes the benchmark prediction—contractionary policy lowers employment, more so where interest-sensitivity is high—while the identification and shift-share literatures explain how that prediction can appear to reverse in practice. Every method I use is disciplined by this body of work: the structural VAR follows Christiano et al. (1999), the projections follow Jordà (2005) under the equivalence established by Plagborg-Møller and Wolf (2021), the shock follows Bauer and Swanson (2023) within the high-frequency tradition of Gürkaynak et al. (2005) and Nakamura and Steinsson (2018), and the cross-sectional design is read through Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022). The contribution of this paper is to hold that cross-sectional design fixed and vary only the source of policy variation, isolating whether the puzzling positive sign reflects spatial aggregation or identification, while recognizing that the information-effect decomposition of Jarociński and Karadi (2020) and shadow-rate measures such as Wu and Xia (2016) mark the natural next steps.

3 Data

County employment is the third-month establishment employment level from the QCEW (Quarterly Census of Employment and Wages), 2002Q1–2019Q4, for ∼2,889 counties; the log level is the outcome. Predetermined exposure is the 2002 county share of employment in construction and durable manufacturing, built from County Business Patterns (CBP), treated as time-invariant (the shift-share "shares"). National macro series are from FRED: the effective federal funds rate (`FEDFUNDS`), total nonfarm payrolls (`PAYEMS`), and the CPI (`CPIAUCSL`), all monthly. The identified monetary shock is the Bauer and Swanson (2023) orthogonalized high-frequency surprise (monthly, summed to quarters for the panel; San Francisco Fed update, covering 1988–2023). Employment and prices enter as $100 \times \Delta \log$; the funds rate enters in levels and first differences.

The county panel and surprise series are aligned at quarterly frequency over 2002–2019 (72 quarters). The national VAR/SVAR use monthly data; the benchmark VAR is estimated on 1976–2008 (a clean, pre-ZLB modern sample) and re-estimated on 2002–2019 to isolate the ZLB's effect. Table 1 reports summary statistics. Employment growth and inflation are mildly left-skewed and leptokurtic, reflecting the 2008–09 collapse; the funds-rate change is sharply non-normal (excess kurtosis 15.9) because it is near-zero throughout the ZLB and moves only in discrete tightening/easing episodes.

Table 1: Summary statistics, 2002–2019 (monthly)

Series	Mean	Std. dev.	Min	Max	Skew.	Kurt.
Fed funds rate (level, %)	1.420	1.581	0.07	5.26	1.226	0.424
Δ Fed funds rate	−0.001	0.141	−0.96	0.25	−3.001	15.883
Employment growth ($\Delta \log \times 100$)	0.068	0.158	−0.62	0.40	−1.998	5.194
Inflation ($\Delta \log$ CPI$\times 100$)	0.175	0.295	−1.786	1.367	−1.430	10.310

Figure 1 plots the three national series with the ZLB window (Dec 2008–Dec 2015) shaded. The funds rate falls to the ZLB in late 2008 and stays there through 2015, so ΔFFR is essentially zero for seven years while employment recovers steadily—the mechanical source of the positive employment-to-tightening comovement documented below. Table 2 reports augmented Dickey–Fuller (ADF) and KPSS tests. Inflation is clearly stationary on both tests. For the funds-rate change and employment growth the picture is the familiar mixed one: the ADF cannot reject a unit root at the data-driven lag (a reflection of its low power on short, persistent series) while KPSS is borderline, and monthly payroll growth carries enough residual seasonality to nudge its KPSS statistic past the 5% line. The funds-rate *level* is non-stationary (ADF $p = 0.46$; KPSS rejects), as expected for a near-integrated policy rate. Following standard practice, all VAR/SVAR variables therefore enter in stationary ($\Delta \log$ or first-differenced) form.

Table 2: Stationarity tests (2002–2019)

Series	ADF p-value	KPSS p-value
Fed funds rate (level)	0.464	0.010
Δ Fed funds rate	0.263	0.100
Employment growth	0.454	0.018
Inflation	0.010	0.100

ADF H_0: unit root. KPSS H_0: stationarity.

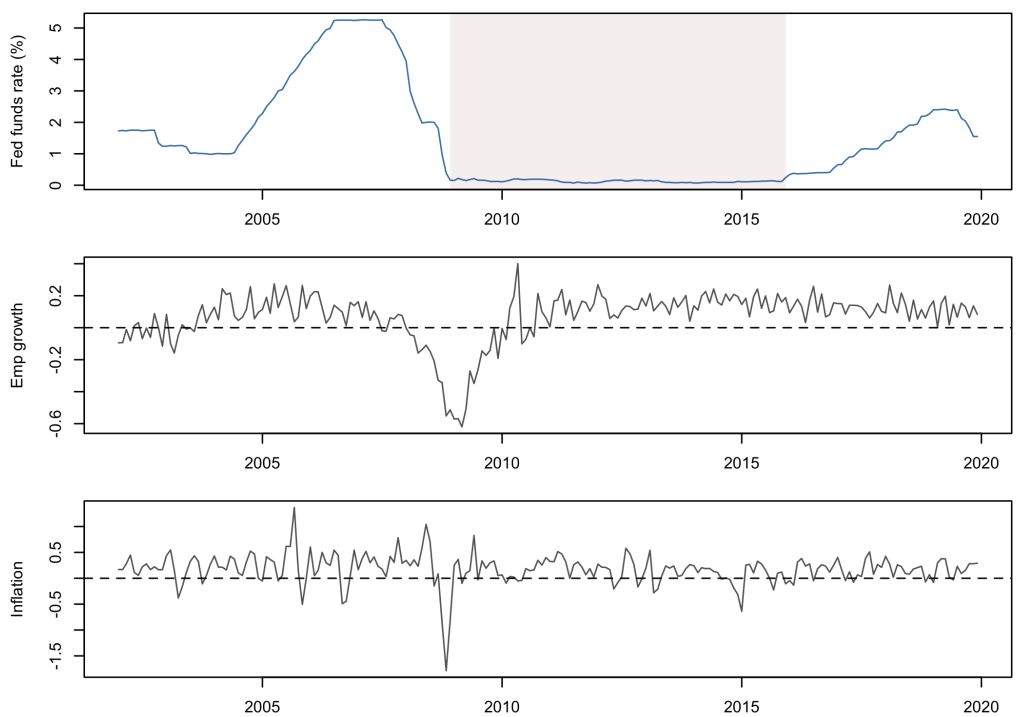

Figure 1: U.S. monetary and labor-market series, 2002–2019. Shaded band marks the zero-lower- bound episode (Dec 2008–Dec 2015), during which ΔFFR is pinned near zero.

4 Empirical Methodology

The analysis applies four of the methods on the ECO 508 list, each chosen to serve a distinct role in adjudicating the puzzle. Throughout, let $\pi_t = 100\,\Delta\log\mathrm{CPI}_t$ denote monthly inflation, $n_t = 100\,\Delta\log\mathrm{PAYEMS}_t$ employment growth, and r_t the federal funds rate with Δr_t its change.

The national benchmark is a reduced-form vector autoregression in $y_t = (\pi_t, n_t, \Delta r_t)'$,

$$y_t = c + \sum_{k=1}^{p} A_k\, y_{t-k} + u_t, \qquad u_t \sim (0, \Sigma_u), \tag{1}$$

with the lag length p chosen by the AIC, SC, and HQ criteria; from it I report Granger-causality tests, forecast-error variance decompositions, and impulse responses with bootstrap confidence bands. To give the policy innovation an economic interpretation, I identify a structural monetary shock recursively in the manner of Christiano et al. (1999), writing $u_t = B\varepsilon_t$ with B the lower-triangular Cholesky factor of Σ_u and the funds rate ordered last,

$$\begin{pmatrix} u_t^{\pi} \\ u_t^{n} \\ u_t^{r} \end{pmatrix} = \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_t^{\pi} \\ \varepsilon_t^{n} \\ \varepsilon_t^{\mathrm{MP}} \end{pmatrix}. \tag{2}$$

Equation (2) is the identifying restriction matrix: the zero restrictions encode the assumption that prices and employment do not respond to the monetary shock $\varepsilon_t^{\mathrm{MP}}$ within the month, while the policy rate may respond to all contemporaneous information. The three structural innovations are a price (supply) shock ε_t^{π}, a real-activity (employment) shock ε_t^{n}, and the monetary policy shock $\varepsilon_t^{\mathrm{MP}}$, with the recursive ordering letting each affect the variables ordered after it contemporaneously.

The cross-sectional design rests on local projections. For the focused panel of major counties i, the Jordà (2005) specification regresses the cumulative change in log employment ℓ on the identified shock at each horizon h,

$$100\big(\ell_{i,t+h} - \ell_{i,t-1}\big) = \alpha_i^h + \delta_t^h + \beta_h\, s_t + \text{controls} + \epsilon_{i,t+h}, \tag{3}$$

where s_t is the Bauer–Swanson orthogonalized surprise entered directly as the shock. The central identification test adds the exposure interaction,

$$100\big(\ell_{i,t+h} - \ell_{i,t-1}\big) = \alpha_i^h + \delta_t^h + \theta_h\,\big(\text{shock}_t \times \widetilde{\exp}_i\big) + \epsilon_{i,t+h}, \tag{4}$$

and is estimated twice: once with $\text{shock}_t = \Delta r_t$, the endogenous funds-rate change that drives the difference-in-differences puzzle, and once with $\text{shock}_t = s_t$, the identified surprise, where $\widetilde{\exp}_i$ is standardized exposure. County and time fixed effects absorb the level of the shock and of exposure, so θ_h is identified purely from the differential response by exposure; I cluster standard errors by quarter, since the identifying variation lives in the seventy-two quarterly shocks (clustering by quarter is the panel analog of a Newey–West HAC correction, absorbing the within-quarter cross-sectional dependence a pure time-series HAC would miss). I use the surprise as a direct shock rather than as an instrument for Δr_t because the quarterly first stage is empty: regressing Δr_t on the orthogonalized surprise yields $F \approx 0$ ($R^2 \approx 0$). Since the orthogonalized surprise does not move the realized rate, the external-instrument approach of Gertler and Karadi (2015) and Stock and Watson (2018) is infeasible here, and using the surprise directly is the standard fallback (Coibion, 2012).

The fourth method is a random forest that forecasts employment growth n_t from a matrix of twelve lags each of n, π, and Δr—thirty-six predictors, all lagged so that no contemporaneous information enters and look-ahead bias is ruled out by construction. I evaluate it out of sample on a rolling, expanding-window basis, refitting each year and benchmarking against a random walk and a seasonal-naive (twelve-month) forecast using RMSE, MAE, and directional accuracy. The evaluation logic is consistent across methods: the VAR and SVAR are judged by the shape of their impulse responses and variance decompositions, the local projections by the horizon-h response and the sign of θ_h, and the random forest by its rolling out-of-sample loss against the naïve benchmarks. All estimation is carried out in the accompanying R script, which reproduces every table and figure end-to-end.

5 Results

5.1 Reduced-form VAR

Lag-selection criteria disagree (Table 3): AIC and FPE choose $p = 12$, SC and HQ choose $p = 3$; I report the AIC system. On the 1976–2008 benchmark, ΔFFR Granger-causes employment ($p = 0.003$; Table 4), and the funds rate accounts for ∼6.5% of the employment forecast-error variance at two years (Table 5). The cumulative employment response to a contractionary shock is negative and hump-shaped, with a trough near −0.38% (Figure 2, left). Re-estimating the identical specification on 2002–2019 *inverts* the response to strongly positive (+2.45% at $h = 12$; Figure 2, right): with the funds rate pinned at the ZLB for seven years, ΔFFR carries almost no policy content and the VAR recovers reverse causation (rates rise as employment recovers).

Table 3: VAR lag selection

Criterion	Lag
AIC	12
SC (BIC)	3
HQ	3
FPE	12

Table 4: Granger causality

Null	p
ΔFFR $\nrightarrow$ emp.	0.003
emp. $\nrightarrow$ ΔFFR	< 0.001

Table 5: FEVD of employment

	π	n	Δr
h=6	.03	.93	.04
h=12	.03	.90	.07
h=24	.08	.86	.07

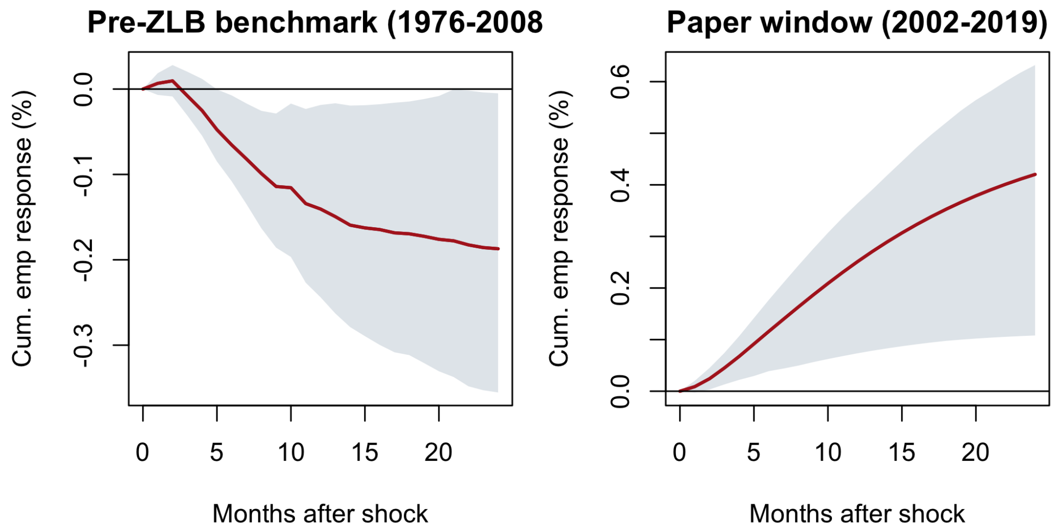

Figure 2: Cumulative employment response to a contractionary monetary shock. Left: pre-ZLB benchmark (1976–2008), the textbook negative response. Right: the 2002–2019 window, where the ZLB inverts the sign. Shaded bands are 95% bootstrap intervals.

5.2 Structural VAR

With the recursive ordering $(\pi, n, \Delta r)$ and the monetary shock identified as the funds-rate innovation, the structural impulse responses (Figure 3) confirm the benchmark: employment falls in a hump-shaped path, troughing near -0.20% (cumulative -0.14% at one year), inflation responds slowly, and the policy rate mean-reverts. The structural and reduced- form responses coincide in sign and shape, as expected when the only identifying content is the recursive timing restriction; the SVAR's value added is the explicit economic interpretation of $\varepsilon^{\mathrm{MP}}$ as an exogenous policy surprise.

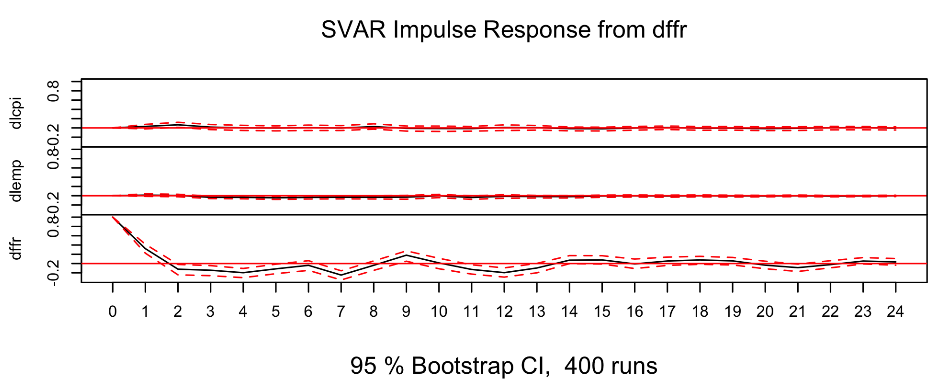

Figure 3: SVAR structural impulse responses to a one-standard-deviation contractionary monetary shock (recursive identification, 1976–2008). Employment (center) declines in a hump-shaped path. Shaded bands are 95% Monte Carlo intervals.

5.3 Local projections: the sign reversal

Figure 4 and Table 6 present the paper's central result. Using the *endogenous* funds-rate change in the shift-share interaction (Eq. 4) reproduces the DiD puzzle: the exposure interaction is positive and strongly significant at every horizon (+0.20 at h=0 rising to +1.25 at h=6, t up to 6.9). Replacing it with the identified Bauer–Swanson surprise *flips the coefficient negative at all horizons* (−0.39 to −2.42), deepening with horizon: the point estimates are negative at every horizon, consistent with theory, though imprecisely estimated. The surprise-based estimates are not individually significant ($|t| \leq 1.2$), reflecting the limited variation in 72 quarterly shocks; the contribution is the *contrast*: the precisely estimated positive coefficient is consistent with an endogeneity artifact, while the identified estimate points negative.

Table 6: Full-panel exposure interaction θ_h: endogenous rate vs. identified surprise

Horizon h (qtrs)	0	1	2	4	6	8
ΔFFR × exp. (endogenous)	0.20**	0.45***	0.73***	1.07***	1.25***	1.16***
(s.e.)	(.08)	(.10)	(.14)	(.21)	(.26)	(.32)
surprise × exp. (identified)	−0.39	−0.73	−1.17	−1.62	−1.92	−2.01
(s.e.)	(.50)	(.71)	(.99)	(1.85)	(2.63)	(2.96)

Outcome: $100(\ell_{i,t+h} - \ell_{i,t-1})$. County + time FE; SE clustered by quarter. Exposure standardized. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

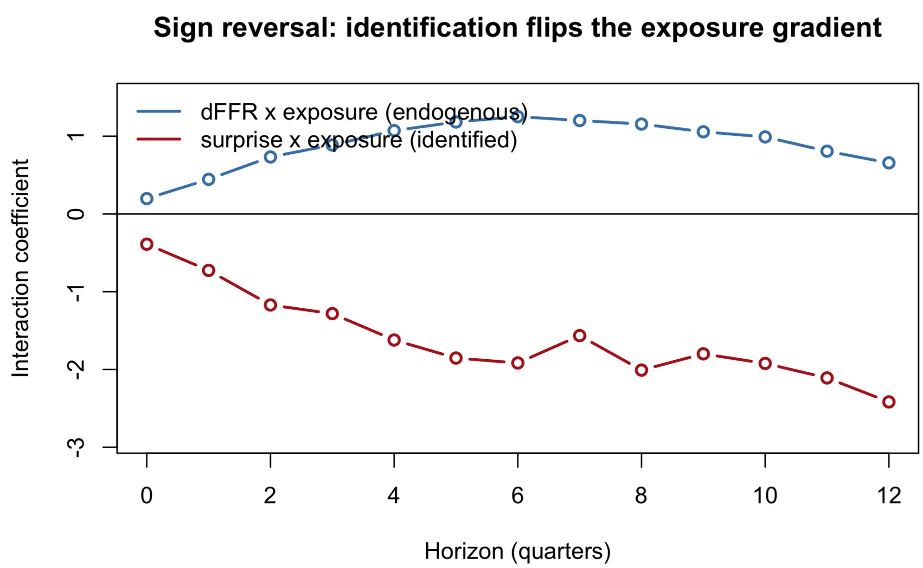

Figure 4: The sign reversal. Holding the shift-share design fixed and swapping only the source of policy variation flips the exposure interaction from significantly positive (endogenous ΔFFR) to negative (identified surprise). Bands are 95% (quarter-clustered).

The focused panel of five major counties (Los Angeles, Cook/Chicago, New York, Harris/Houston, Maricopa/Phoenix), following the proposal feedback to narrow the LP, gives a consistent dynamic picture: the pooled employment response to an identified tightening surprise is negative and deepening to roughly −8% over three years (Figure 5), again imprecise. Reassuringly, the LP and VAR employment responses share the same hump-shaped contraction once placed on a common scale (Figure 6); since Plagborg-Møller and Wolf (2021) prove LP and VAR target the same population responses, this agreement is expected, with cross-method differences reflecting shock measurement rather than the estimator (Coibion, 2012).

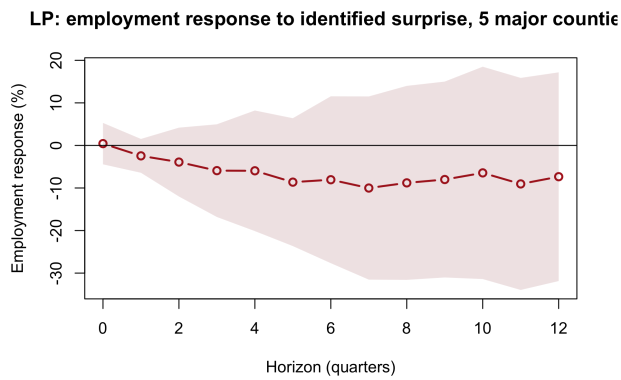

Figure 5: LP: employment response to an identified surprise, five major counties.

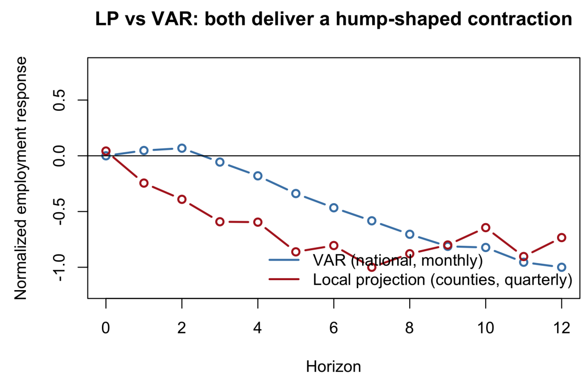

Figure 6: LP vs. VAR employment responses (normalized): both hump-shaped contractions.

A caution on the cross-section. The exposure gradient is not robust to county size. Among the 20 largest counties split by exposure (so size is held roughly fixed), high- and low-exposure metros contract *similarly* after an identified shock (difference $|t| < 1.3$ at all horizons): extreme-exposure counties are small, specialized places, so a naïve high-vs- low split confounds exposure with size and diversification. The honest reading—reinforcing the thesis—is that *identification*, not a cross-sectional exposure channel, drives the sign: clean shocks lower employment broadly, and the DiD's positive interaction was an endogeneity artifact.

5.4 Random forest

Out-of-sample, the random forest attains a lower average RMSE (0.059) than the random-walk (0.081) and seasonal-naive (0.084) benchmarks across rolling test years (Table 7). But this is not forecasting skill: the correlation between RF forecasts and realized employment growth is −0.03, and over the 2014–2019 expansion the RF essentially predicts the stable positive mean (Figure 7, right). The RMSE "win" is variance shrinkage toward the conditional mean in a low-volatility expansion, not genuine predictability; directional accuracy is trivially high only because growth was positive in every test month. Variable importance (Figure 7, left) loads almost entirely on own lags of employment growth, and the partial dependence on the first lag (Figure 8) is nearly flat—consistent with weak short-horizon predictability of the aggregate. This tempers the optimism of Medeiros et al. (2021): random forests help for some series and horizons, but here they add no skill beyond a naïve benchmark.

Table 7: Random forest rolling-origin out-of-sample RMSE and MAE

Test year	RF RMSE	RW RMSE	Seas.-naive RMSE	RF MAE	RW MAE
2014	0.065	0.063	0.081	0.055	0.055
2015	0.054	0.083	0.074	0.044	0.074
2016	0.064	0.099	0.096	0.054	0.088
2017	0.036	0.037	0.078	0.030	0.029
2018	0.068	0.100	0.066	0.055	0.090
2019	0.066	0.103	0.108	0.055	0.091
Mean	**0.059**	0.081	0.084	**0.049**	0.071

Despite the RMSE/MAE edge, corr(forecast, actual) = -0.03: variance shrinkage, not skill.

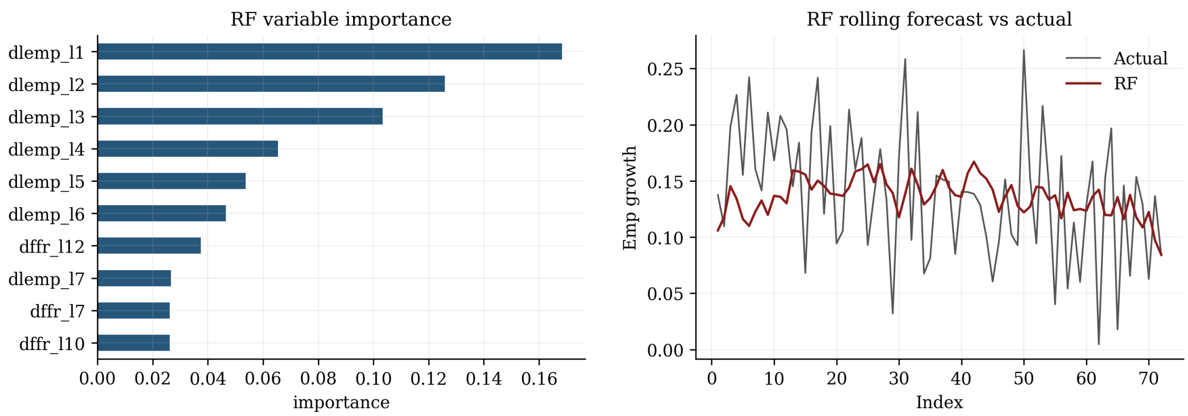

Figure 7: Random forest. Left: variable importance loads on own employment-growth lags. Right: rolling forecasts track the expansion mean but not the fluctuations (corr = -0.03).

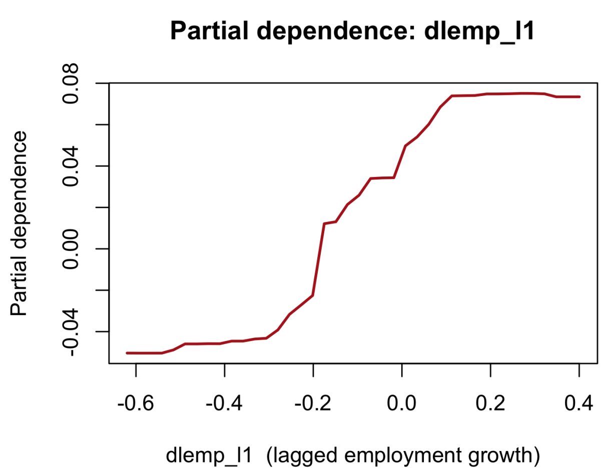

Figure 8: Partial dependence of forecast employment growth on its first lag—nearly flat.

6 Conclusion

This paper adjudicated between two explanations for a counterintuitive positive employment response to monetary tightening in a county shift-share design. The evidence points to **identification, not aggregation**. Reduced-form and structural VARs reproduce the textbook contractionary employment response on a clean pre-ZLB sample, and the same VAR inverts over 2002–2019 only because the zero lower bound strips the funds rate of policy content. Most directly, holding the shift-share design fixed and replacing the endogenous funds-rate change with an identified Bauer–Swanson surprise flips the exposure interaction from significantly positive to negative at every horizon. The most plausible reading is that the positive sign is an endogeneity artifact and the underlying effect is contractionary—though, given the imprecision of the surprise-based estimates, this is strongly suggestive rather than definitively established.

Relative to Carlino and DeFina (1998) and the identification literature (Christiano et al., 1999; Coibion, 2012), the paper's contribution is to show concretely how endogeneity can manufacture a sign reversal inside an otherwise standard regional design, and how a high-frequency external shock (Bauer and Swanson, 2023) restores the expected contractionary sign. The finding is therefore as much methodological as substantive: it cautions that shift-share designs built on the realized policy rate can mislead precisely where the policy rate is most endogenous.

These conclusions come with real limits. The surprise-based estimates are imprecise, because seventy-two quarters of monetary surprises provide limited identifying variation, so the negative interaction is directional rather than statistically significant, and an instrumental-variables projection is infeasible because the orthogonalized surprise does not move the realized quarterly rate. The cross-sectional exposure gradient, in turn, is confounded with county size and does not survive once size is held fixed. A deeper alternative I cannot fully rule out is the central-bank information effect (Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020): if the surprises blend pure policy and information shocks, part of the interaction may reflect favorable news rather than endogeneity, and decomposing the two is the priority extension. Other natural steps include a Wu–Xia shadow rate (Wu and Xia, 2016) to recover policy content through the zero lower bound, a longer surprise sample, and Driscoll–Kraay or wild-cluster inference to sharpen the panel projection. The random-forest exercise, meanwhile, stands as its own cautionary note: a lower RMSE need not mean genuine predictive skill.

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A Appendix: R Code

The complete, commented R script reproducing every table and figure is submitted separately as `Dalis_Abdallah_ECO508_Code.R.txt`. It loads the QCEW county panel and CBP exposure shares, downloads the FRED series and the Bauer–Swanson surprises, and runs the VAR, SVAR, local projections, and random forest end-to-end.